

TEMPEST: Soil Gas Well

July 2025 Samples

2026-09-24

Contents

1	Assess standard curves CH₄	4
2	Assess standard curves CO₂	6
3	Assess standard curves N₂O	8
4	Aanlayze the CH₄ Check Standards	14
5	Aanlayze the CO₂ Check Standards	15
6	Aanlayze the N₂O Check Standards	16
7	Analyze the CH₄ Blanks	17
8	Analyze the CO₂ Blanks	18
9	Analyze the N₂O Blanks	19
10	Check Range - CH₄	21
11	Check Range - CO₂	22
12	Check Range - N₂O	23
13	Calculate gas in water - CH₄	24
14	Calculate gas in water - CO₂	25
15	Calculate gas in water - N₂O	26
16	Merge samples with Metadata & check all samples are present	26

##Run Information

```
cat("Run Information: Brenden Blakley ") #lets you know what section you're in
```

Run Information: Brenden Blakley

```
#set the run date & user name
```

```
sample_year <- "2025"
```

```
sample_month <- "July"
```

```
user <- "Brenden Blakley"
```

```
#identify the files you want to read in
```

```
#read in as a list to accommodate multiple runs in a month (.txt file(s))
```

```
GHG_files <- c("Raw Data/20260827_TMP2025_July.txt")
```

```
#Metadata file path
```

```
Metadata_path = "Metadata/TEMPEST_SGAS_202507_ALL_WORKING.csv"
```

```
# Define the file path for QAQC log file - NO Need to change just check year
```

```
log_path <- "GC_QAQC_Log.csv"
```

```
check_log_path <- "GC_Check_QAQC_Log.csv"
```

```
# Define final path for data be sure to change year and month and run number
```

```
final_path <- "Processed Data/TMP2025_July.csv"
```

```
##Standard Type ("Actual" or "Half")
```

```
#LTREB, SMARTX, and GENX PW Samples use "Half", others should use "Actual"
```

```
Standard_Type <- "Actual"
```

```
#Did Some Samples require manual area retrieval?
```

```
##List Samples and their areas for each analyte ("CH4", "CO2", or "N2O")
```

```
Manual_Areas <- data.frame(
```

```
  Sample = c("NA"),
```

```
  Area = c("NA"),
```

```
  Analyte = c("NA")
```

```
)
```

```
#Gas in Water Calculations
```

```
#Are All Samples Gas in Water? ("YES" or "NO")
```

```
GIW <- "NO"
```

```
#Are some samples Gas in Water?
```

```
##List Sample IDs
```

```
GIW_Samples <- c()
```

```
# "TempC" column value in Celsius (equilibration temperature) and volume of gas (in liters) in syringe
```

```
TempC = 25 #Equilibration Temperature (Celsius)
```

```
volume_g = 0.015 #volume of gas (in liters) in syringe when equilibrated
```

```
volume_w = 0.015 #volume of water (in liters) in syringe when equilibrated (should be equal to volume_g)
```

```
#record any notes about the run or anything other info here:
```

```

run_notes <- ""

#Collection_Dates = "Raw Data/Sample_Collection_Dates_2024.csv"

cat(run_notes)

##Set Up
##Read in metadata and create similar sample IDs for matching to samples
##Read In Data Files
#Get Location of Each Sample in Main Data Frame
#Turn Each Sample into it's own Data Frame
#Get Only Area Tables and Retrieve Info
##QAQC Peak Count and Retention Time

## [1] "If there are too many peaks there is likely a problem with the sample or run and the chromatogr

## [1] "FID: Blank1; 0"          "FID: Blank2; 0"          "FID: Oppm; 0"
## [4] "FID: Std_10K_3mL; 1"      "FID: Std_10K_6mL; 1"    "FID: Std_10K_12mL; 1"
## [7] "FID: Std_10K_24mL; 1"      "FID: Blank3; 0"         "FID: Blank4; 1"
## [10] "FID: 2025_July_8; 3"       "FID: Blank5; 0"         "FID: Blank6; 0"
## [13] "FID: 2025_July_29; 3"      "FID: Blank7; 0"         "FID: 2025_July_33; 3"
## [16] "FID: Blank8; 1"           "FID: Blank9; 0"         "FID: Blank10; 0"

## [1] "ECD: Blank1; 0"          "ECD: Oppm; 0"          "ECD: Std_10K_3mL; 0"
## [4] "ECD: Std_10K_6mL; 0"      "ECD: Std_10K_12mL; 0"  "ECD: Std_10K_24mL; 0"
## [7] "ECD: Blank3; 0"          "ECD: Blank4; 0"         "ECD: 2025_July_10; 2"
## [10] "ECD: Blank5; 0"          "ECD: 2025_July_15; 2"  "ECD: Blank6; 0"
## [13] "ECD: Blank7; 0"          "ECD: Blank8; 0"         "ECD: Blank9; 0"
## [16] "ECD: Blank10; 0"

## [1] "If RTime is wrong check that correct number of peaks were found, if not a peak may have been mi

## [1] "CH4:  "

## [1] "CO2:  "

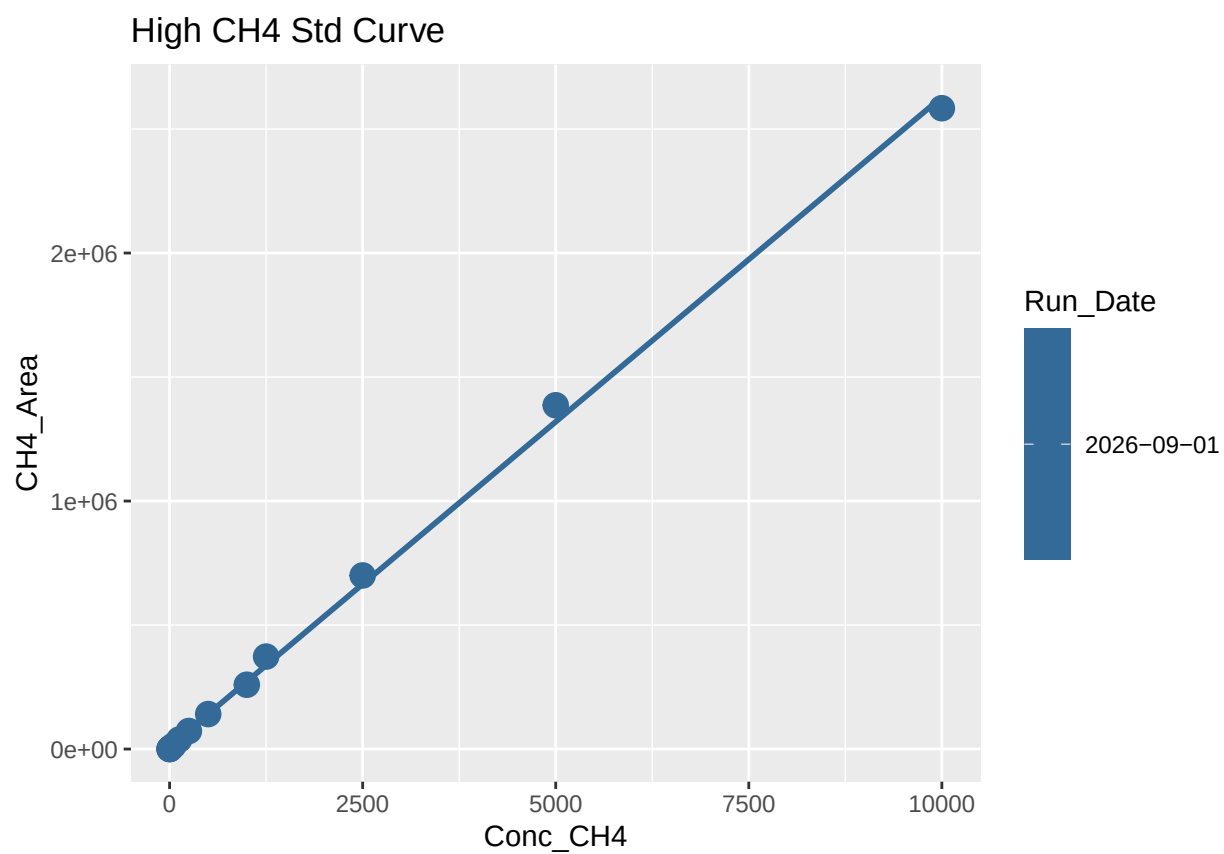
## [1] "N2O: Blank2; 2.765"        "N2O: 2025_July_10; 2.728"
## [3] "N2O: 2025_July_15; 2.758"

##Format Data
##Add Standard Concentrations
##Finalize Data

```

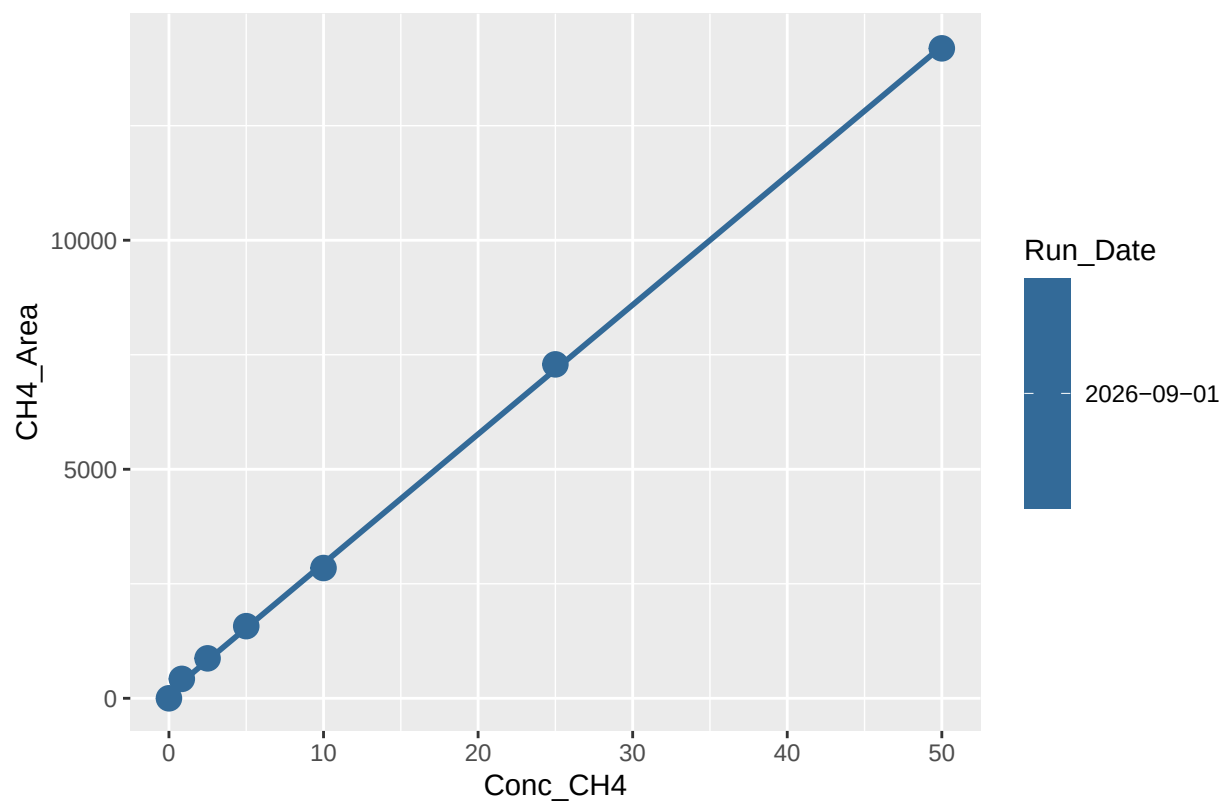
1 Assess standard curves CH4

```
## 'geom_smooth()' using formula = 'y ~ x'
```

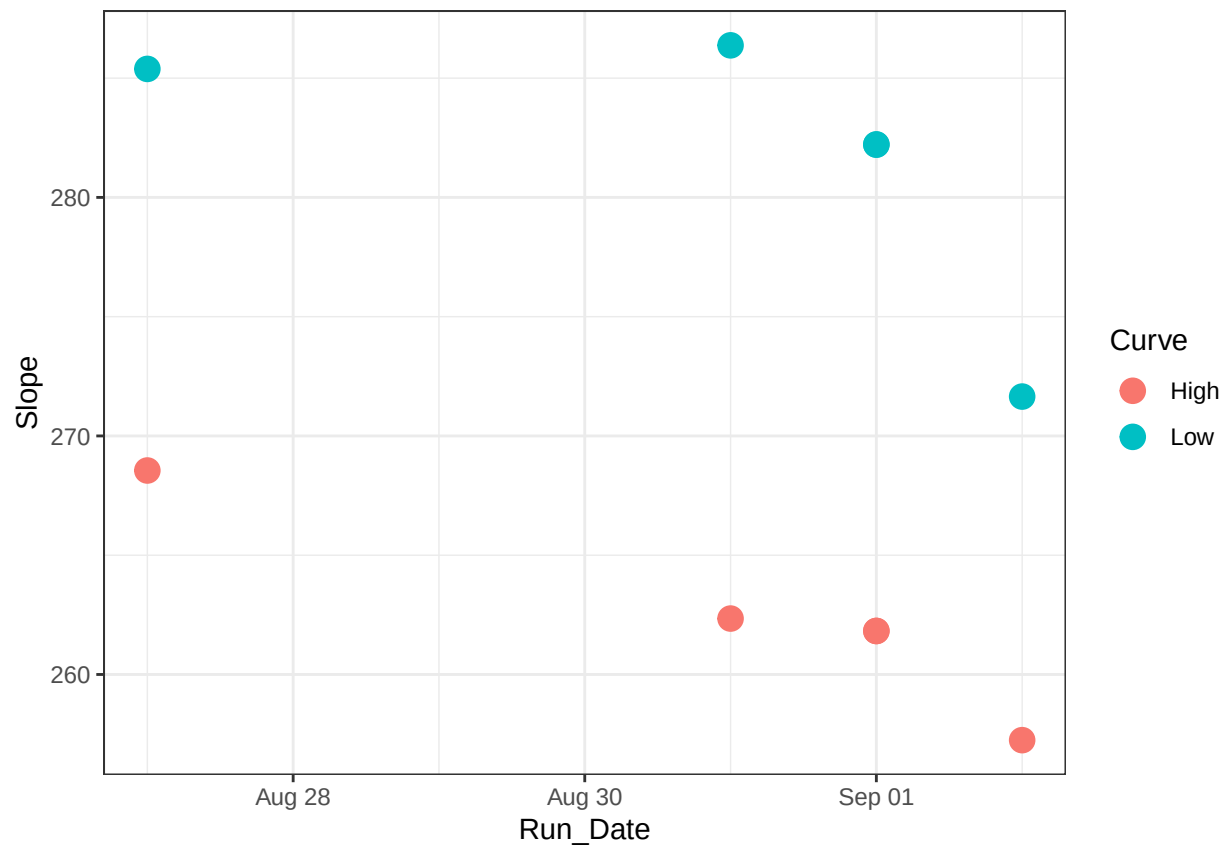


```
## 'geom_smooth()' using formula = 'y ~ x'
```

Low CH4 Std Curve



##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

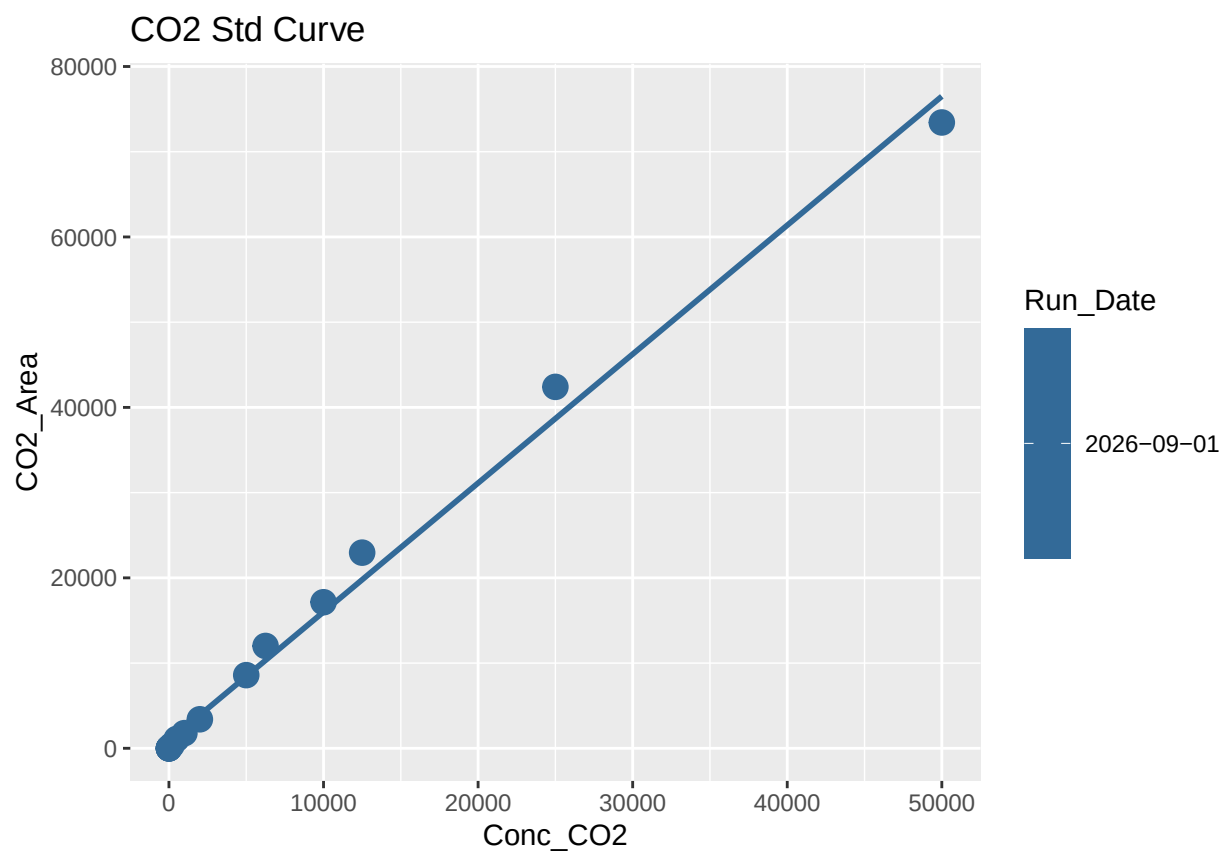


```
## [1] "High_R2 is Good-2026-09-01"
```

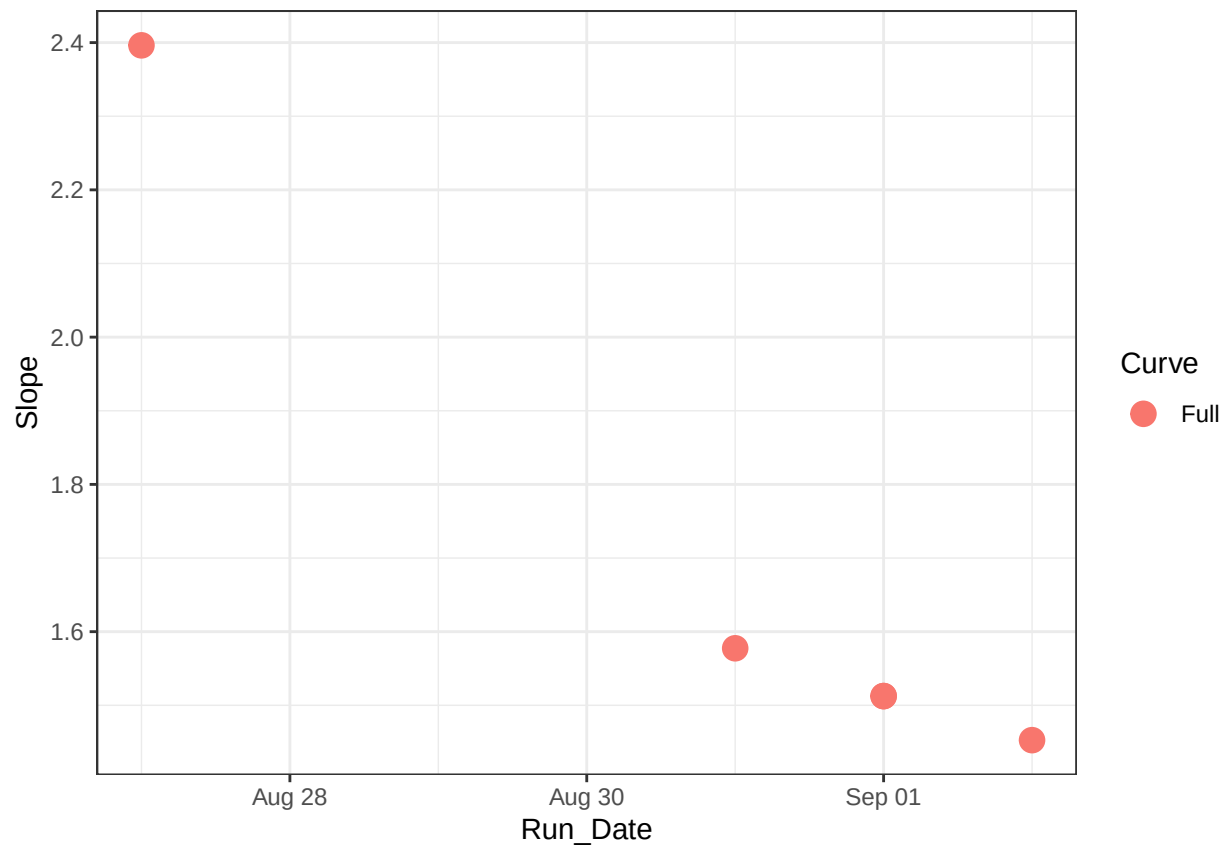
```
## [1] "Low_R2 is Good-2026-09-01"
```

2 Assess standard curves CO2

```
## 'geom_smooth()' using formula = 'y ~ x'
```



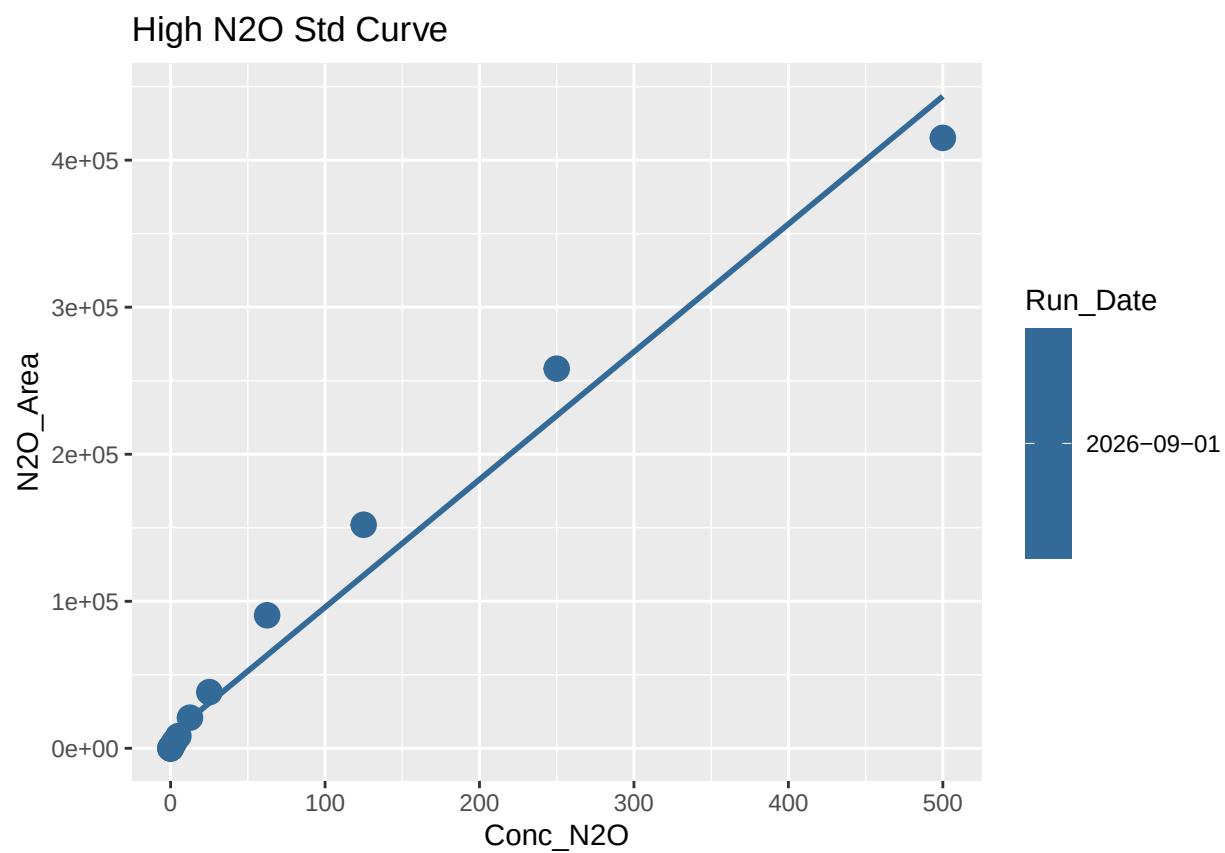
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4



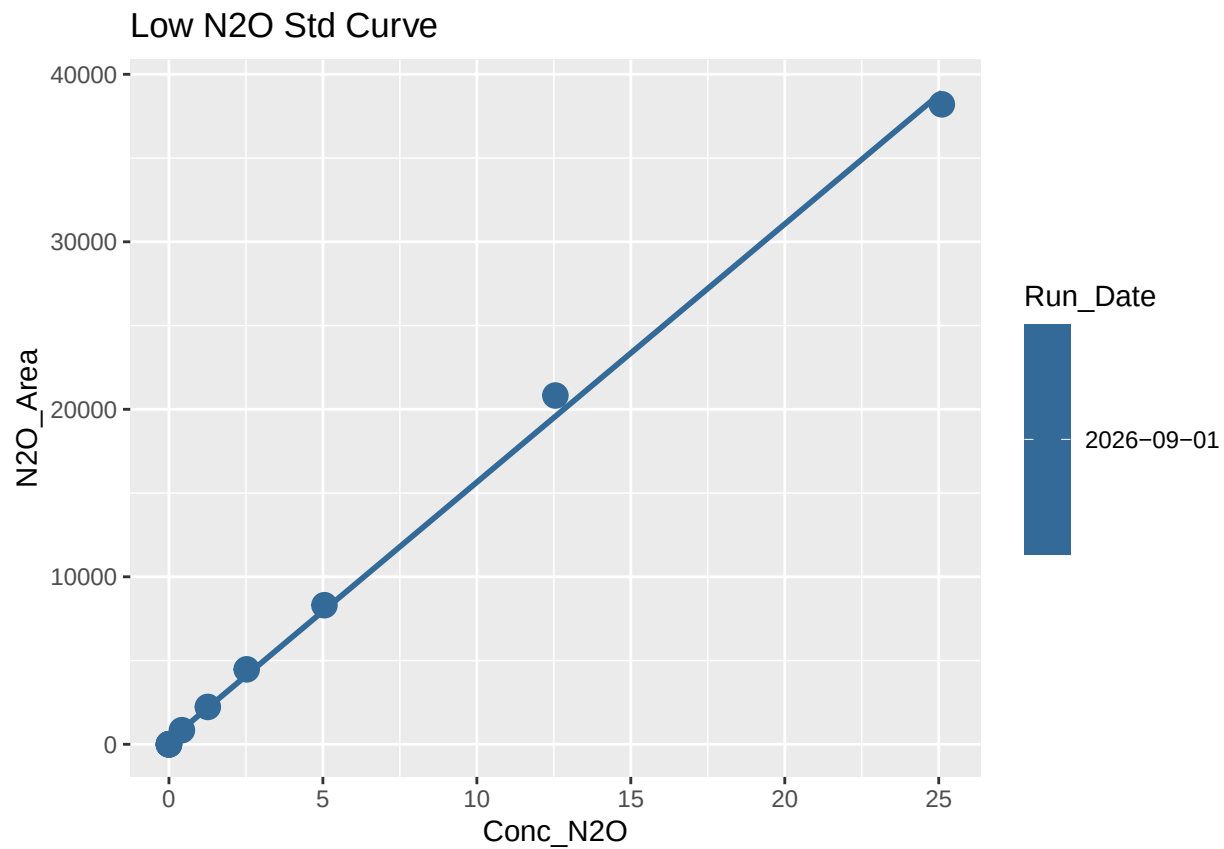
```
## [1] "C02_R2 is Good-2026-09-01"
```

3 Assess standard curves N2O

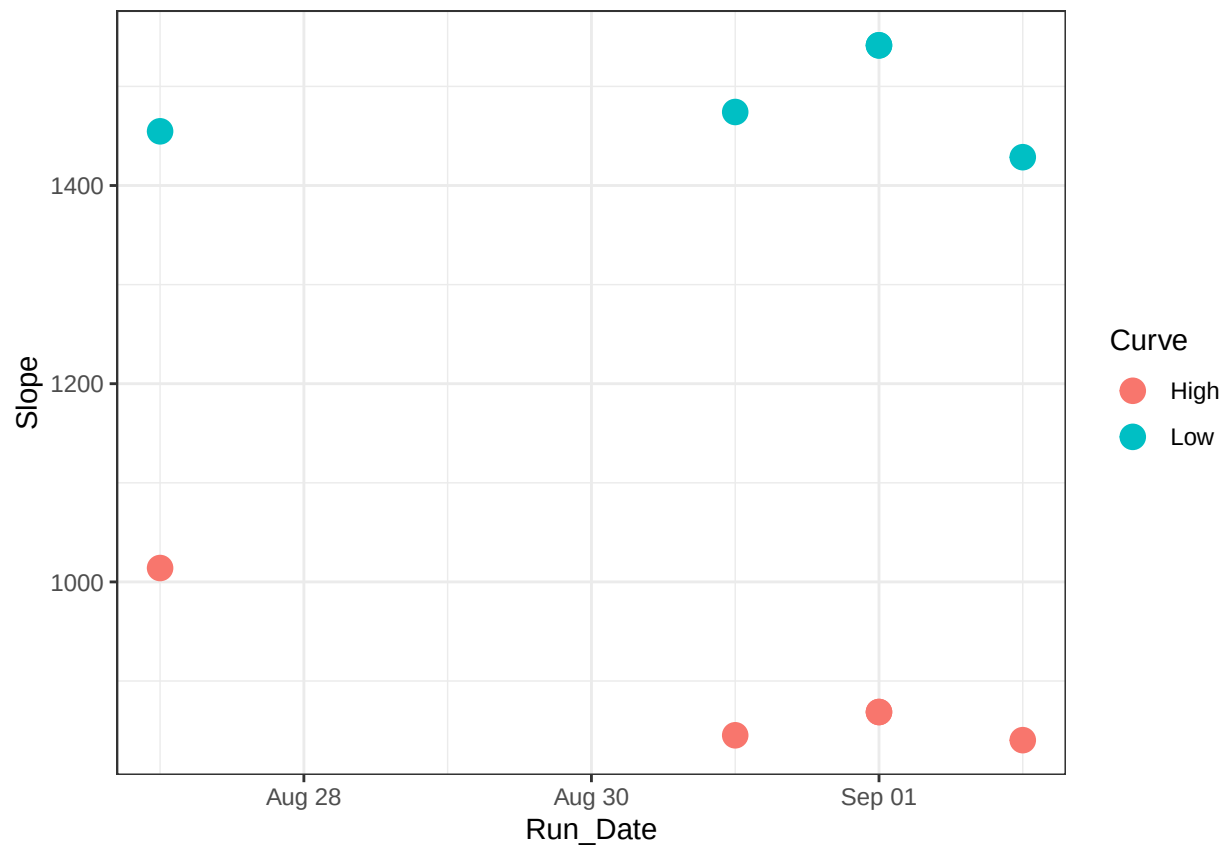
```
## 'geom_smooth()' using formula = 'y ~ x'
```



```
## 'geom_smooth()' using formula = 'y ~ x'
```



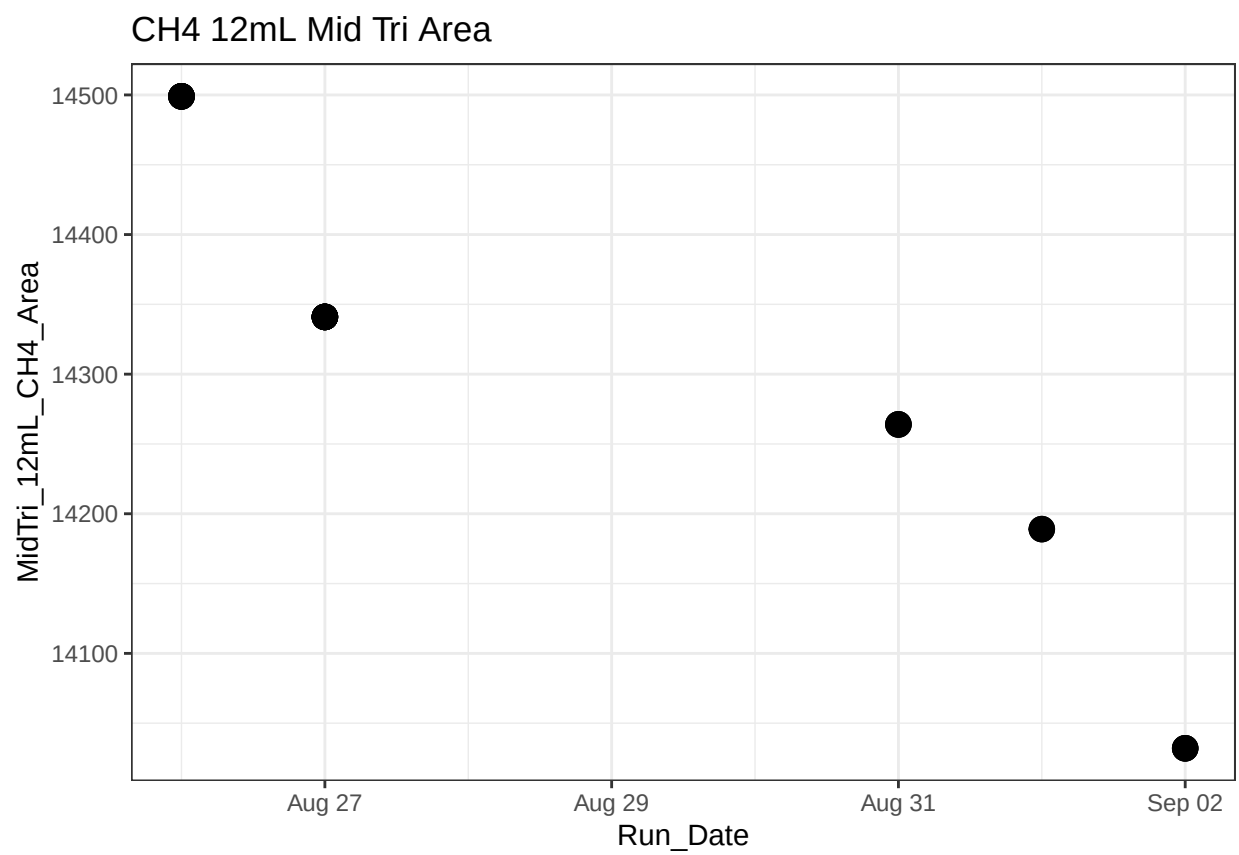
##	Run_Date	Curve	R2	Slope	Intercept	Analyte
## 1	2026-08-27	High	0.9986559	268.551423	9535.1095	CH4
## 2	2026-08-27	Low	0.9995601	285.387517	115.8345	CH4
## 3	2026-08-27	Full	0.9984309	2.396263	254.4449	CO2
## 4	2026-08-27	High	0.9889439	1014.018375	3704.2202	N2O
## 5	2026-08-27	Low	0.9986226	1454.680571	214.6734	N2O
## 6	2026-08-31	High	0.9987205	262.344419	9865.8585	CH4

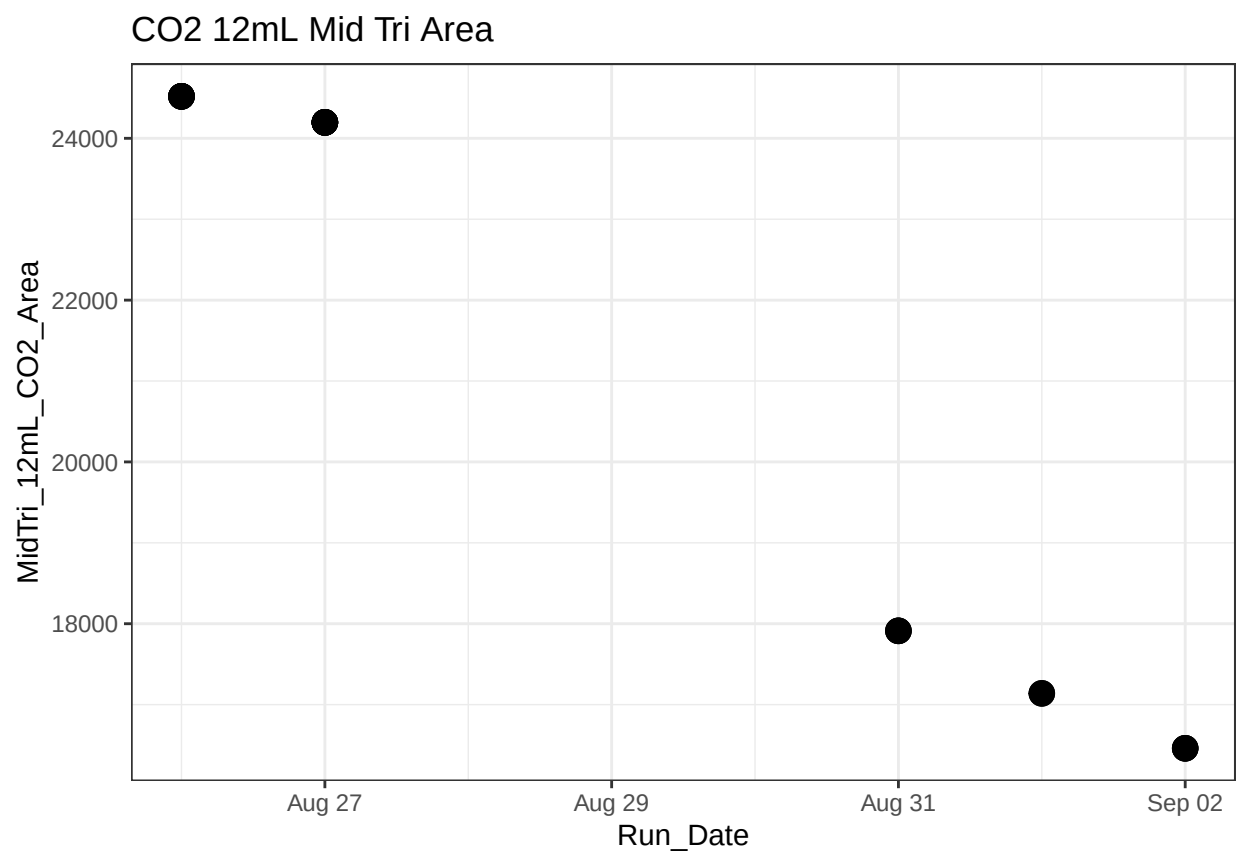


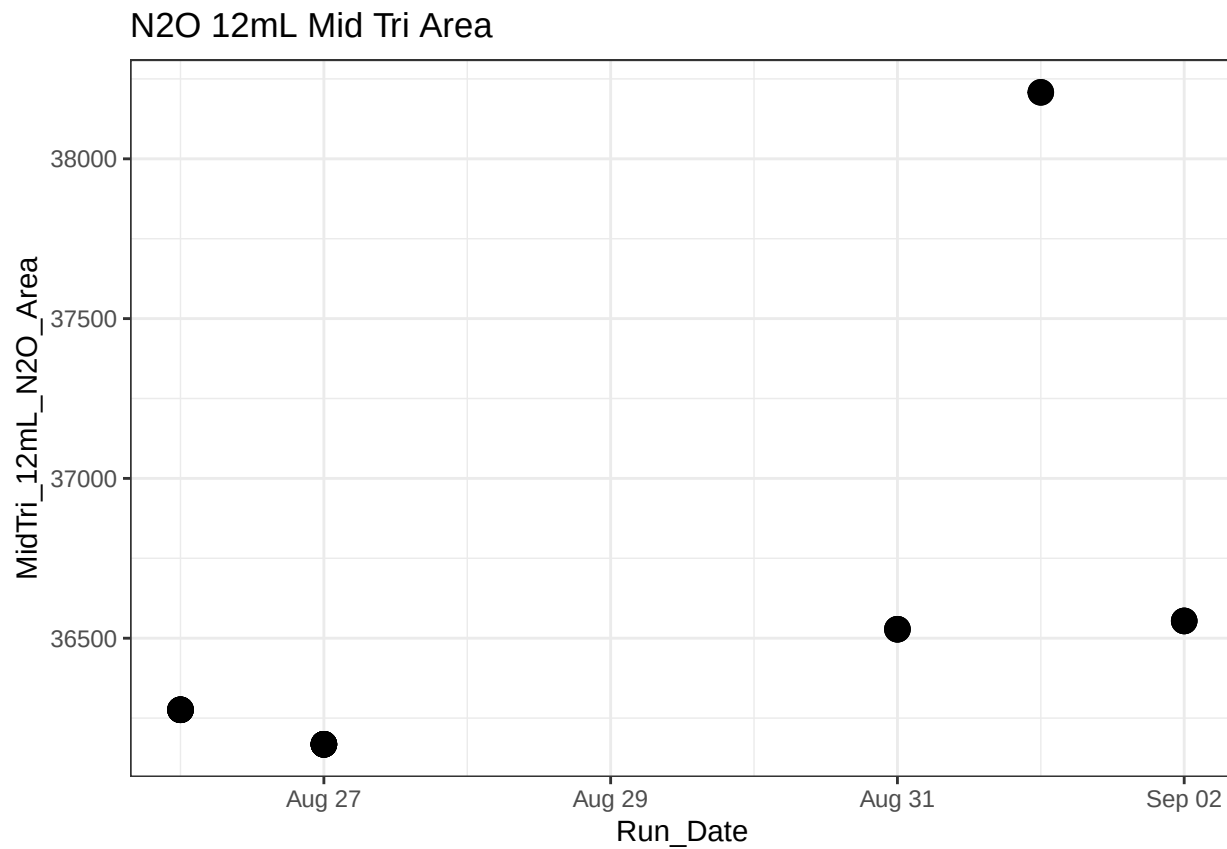
```
## [1] "High_R2 is Bad-2026-09-01"
```

```
## [1] "Low_R2 is Good-2026-09-01"
```

```
#Plot Check Areas over Time
```







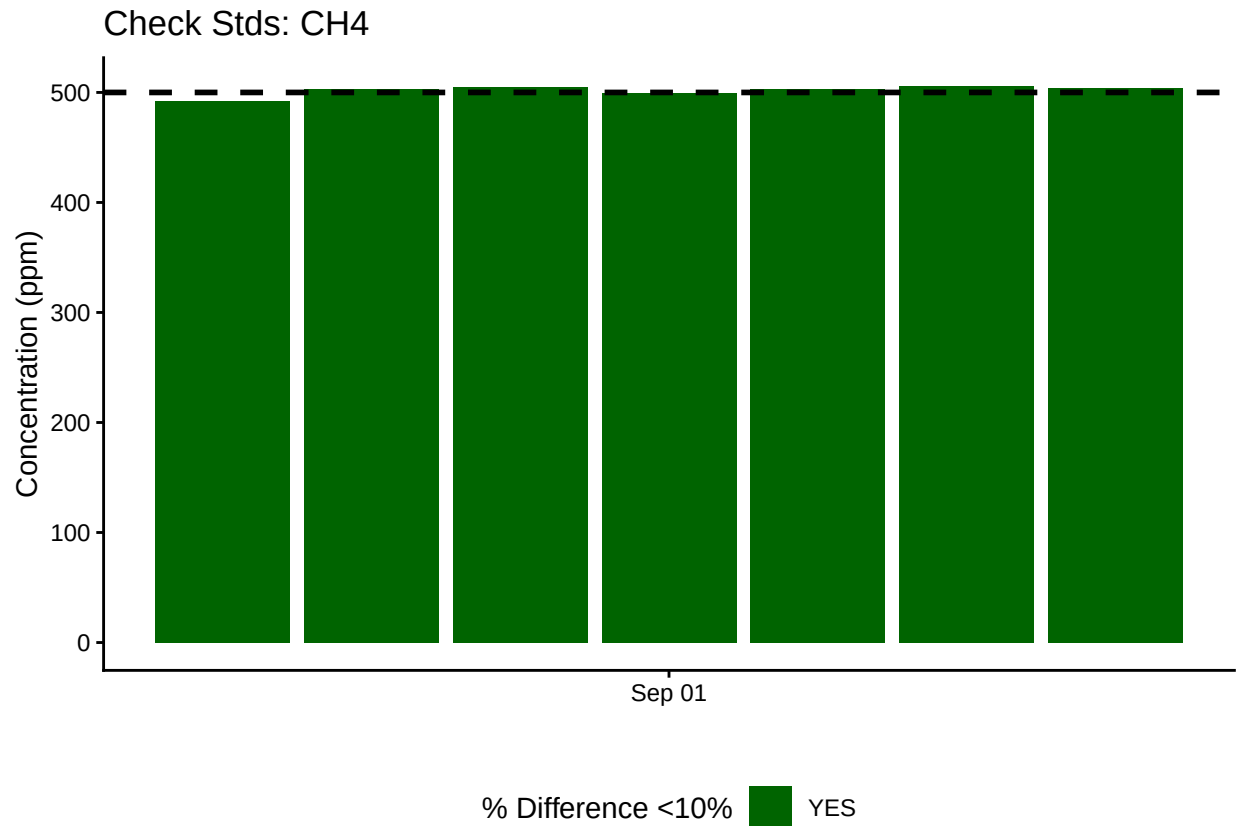
3.1 Now calculate the CH₄ in ppm

3.2 Now calculate the CO₂ in ppm

3.3 Now calculate the N₂O in ppm

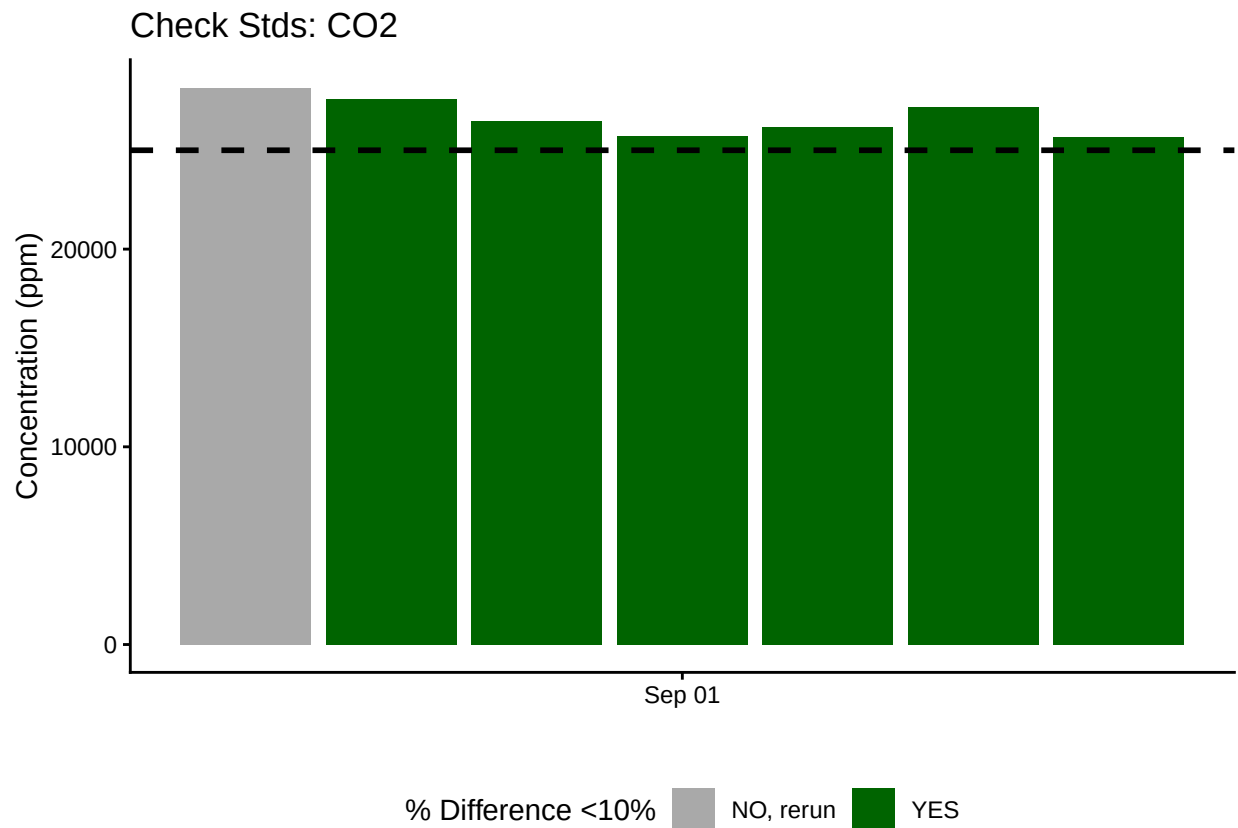
4 Aanlayze the CH₄ Check Standards

>60% of CH₄ Check Standards are within range of expected concentration - PROCEED



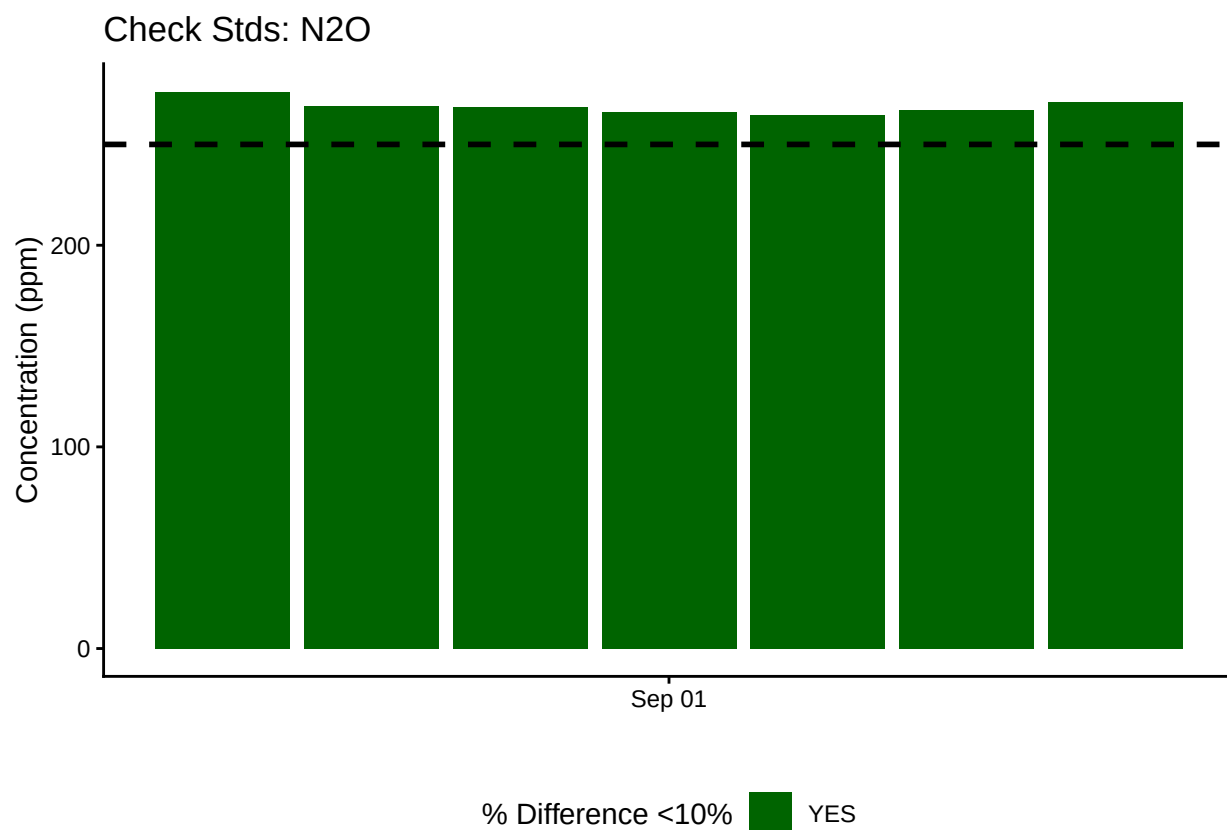
5 Aanlayze the CO2 Check Standards

>60% of CO2 Check Standards are within range of expected concentration - PROCEED



6 Aanlayze the N2O Check Standards

>60% of N2O Check Standards are within range of expected concentration - PROCEED



7 Analyze the CH₄ Blanks

Assess CH₄ Blanks

>60% of CH₄ Blanks are below the lower 25% quartile of samples - PROCEED

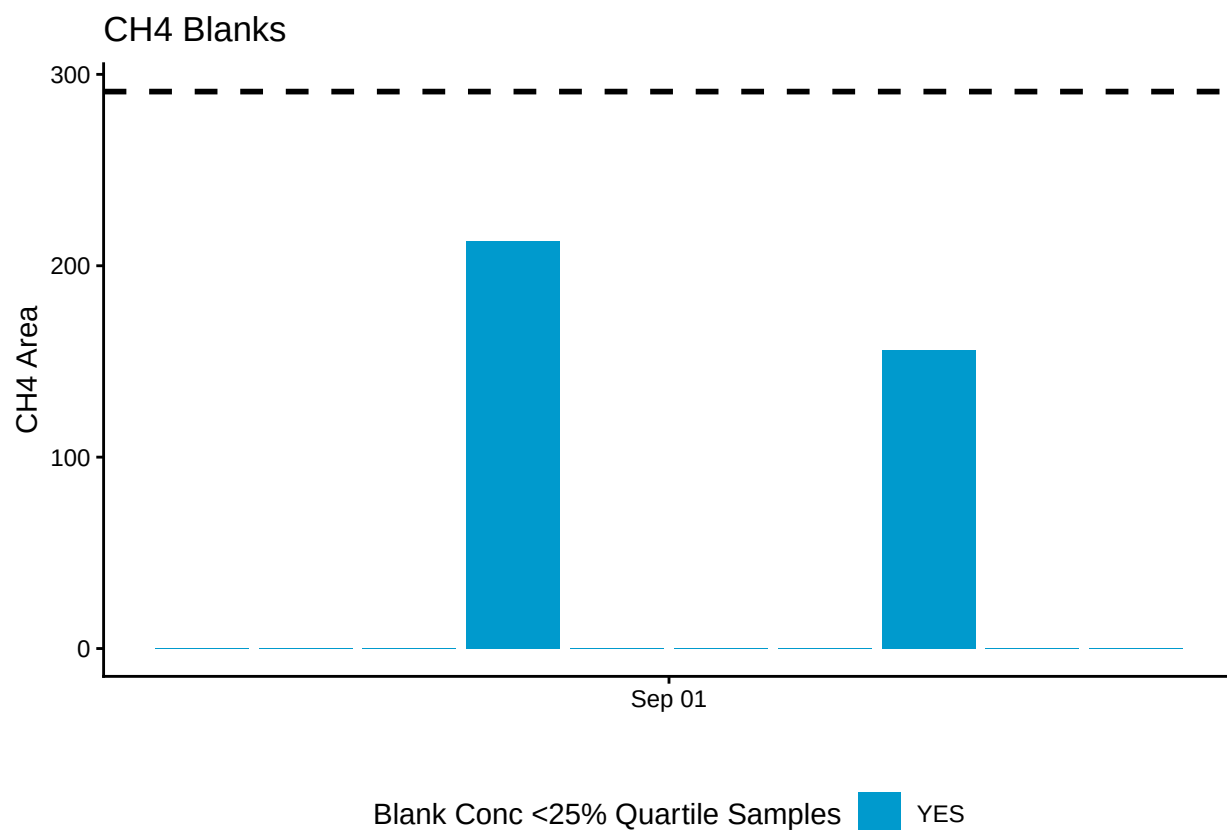


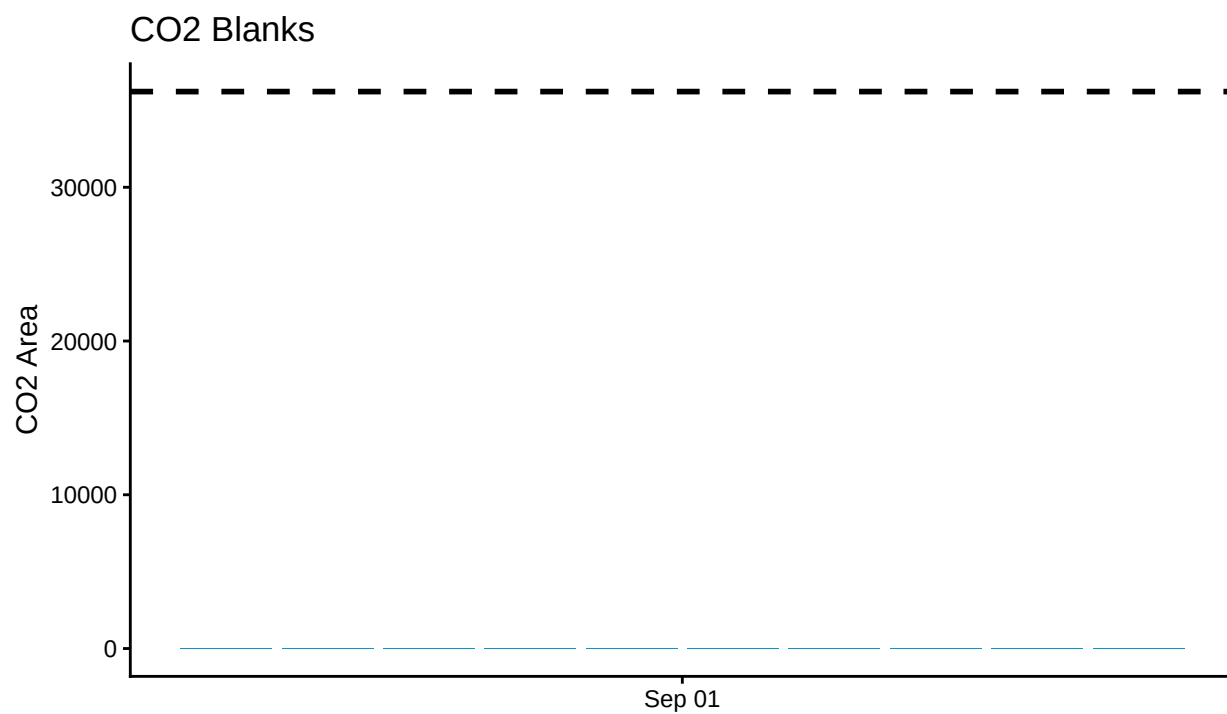
Table 1: Mean Area of Blanks

Test	Blank_Mean_Area
CH4	36.9

8 Analyze the CO2 Blanks

```
## Assess CO2 Blanks
```

```
## >60% of CO2 Blanks are below the lower 25% quartile of samples - PROCEED
```



Blank Conc <25% Quartile Samples ☒ YES

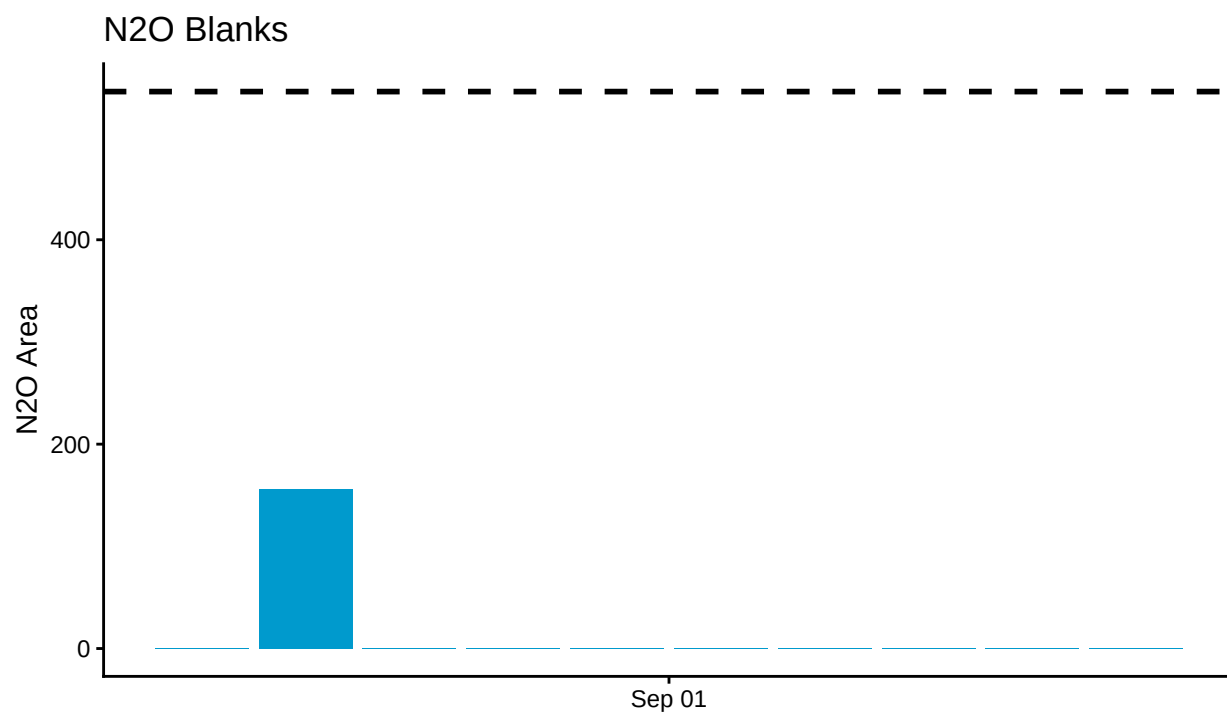
Table 2: Mean Area of Blanks

Test	Blank_Mean_Area
CO2	0

9 Analyze the N2O Blanks

Assess N2O Blanks

>60% of N2O Blanks are below the lower 25% quartile of samples - PROCEED

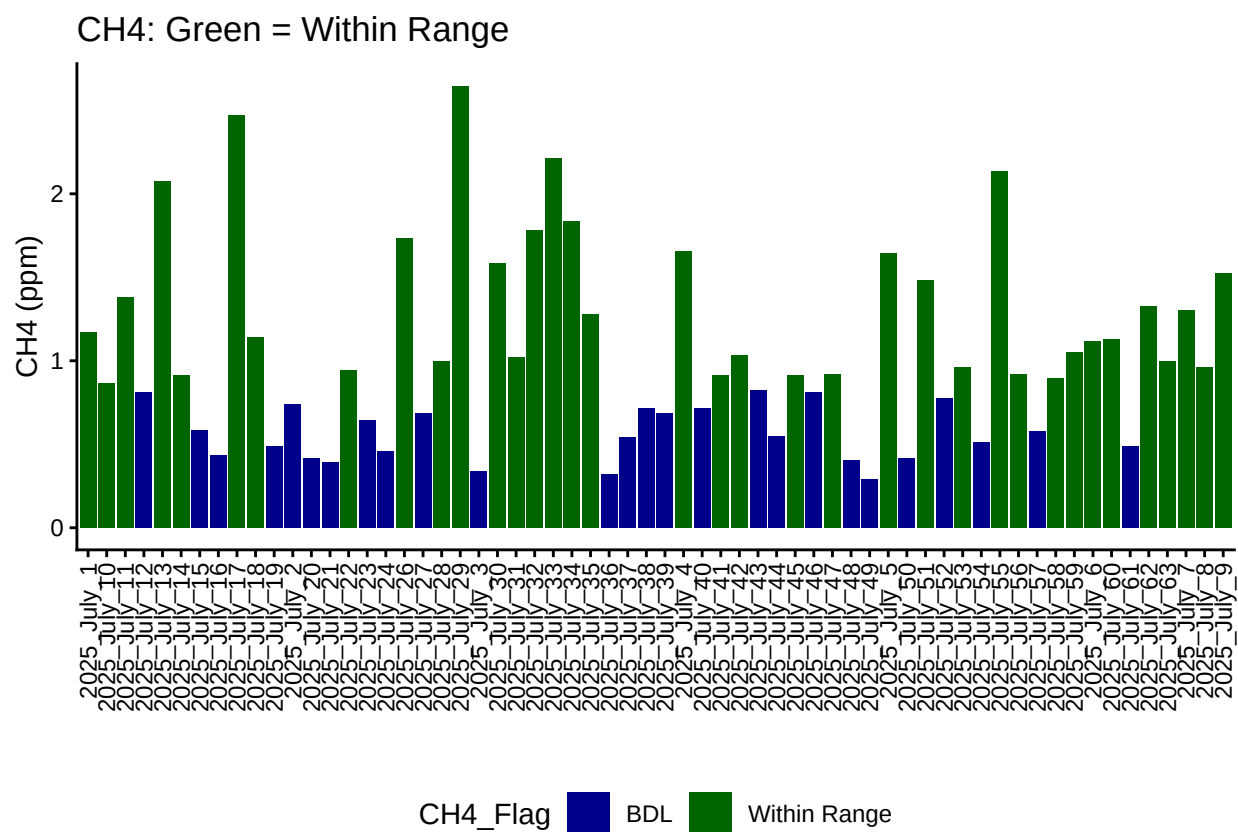


Blank Conc <25% Quartile Samples ■ YES

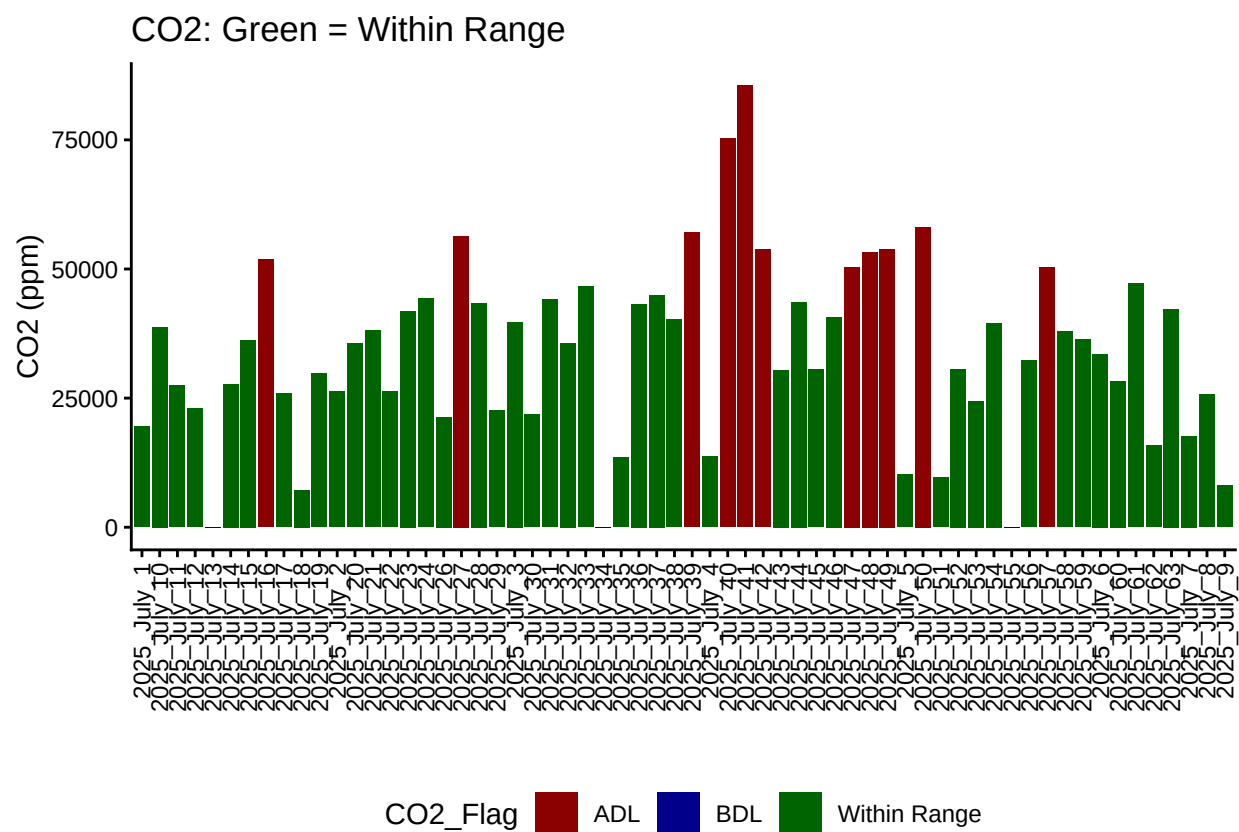
Table 3: Mean Area of Blanks

Test	Blank_Mean_Area
N2O	15.6

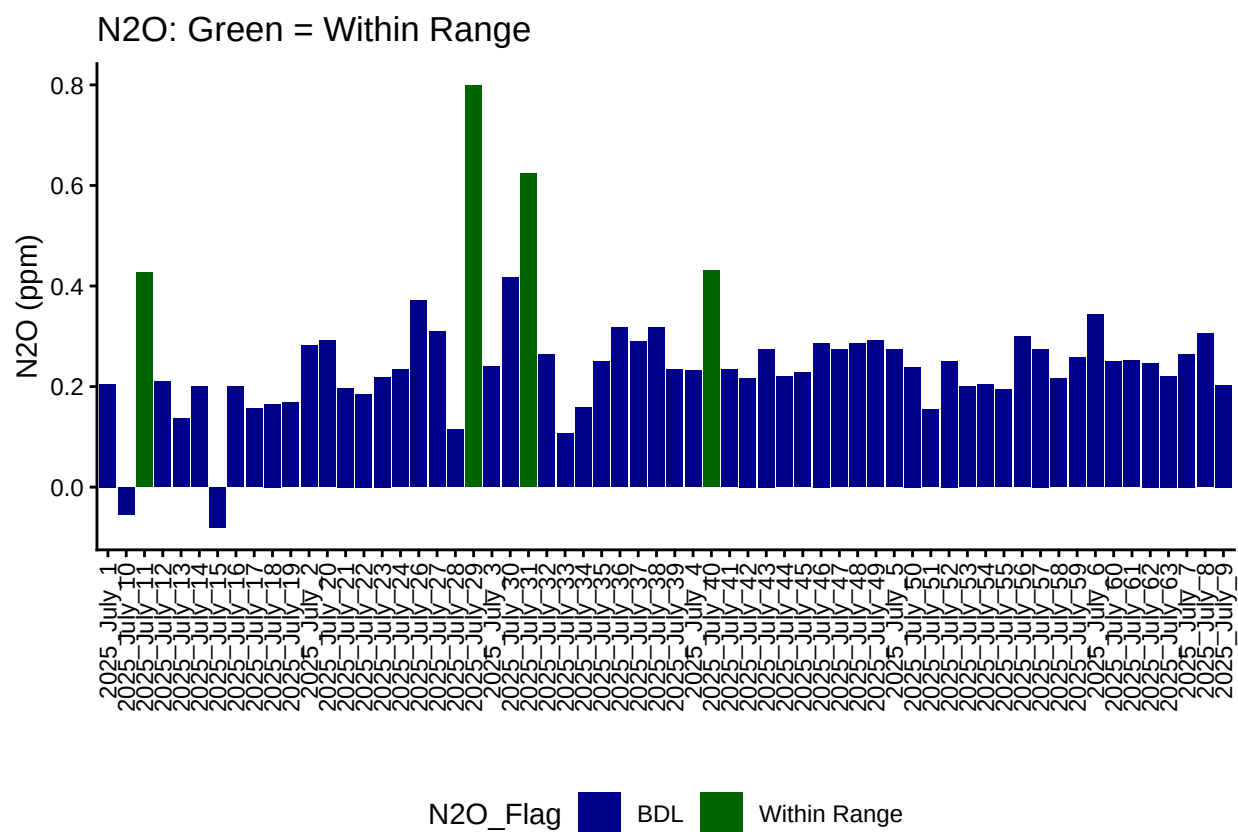
10 Check Range - CH4



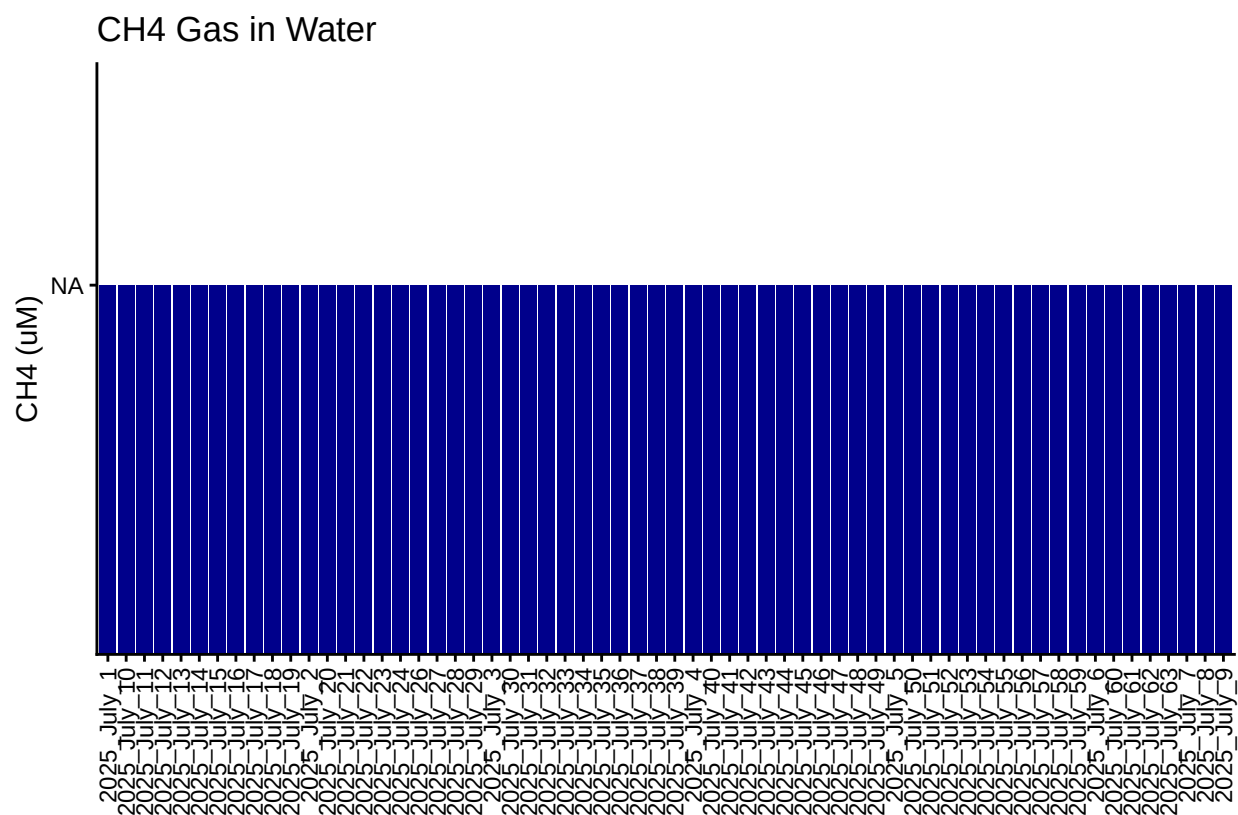
11 Check Range - CO2



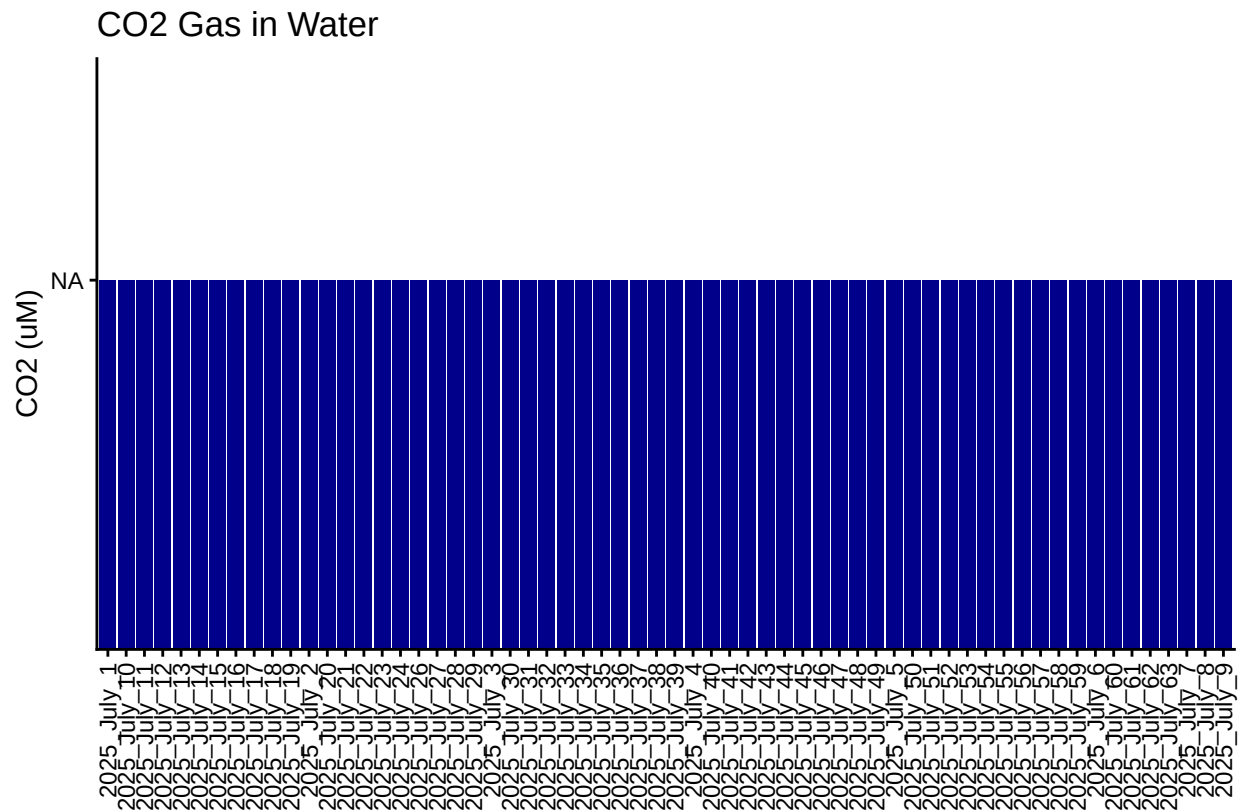
12 Check Range - N2O



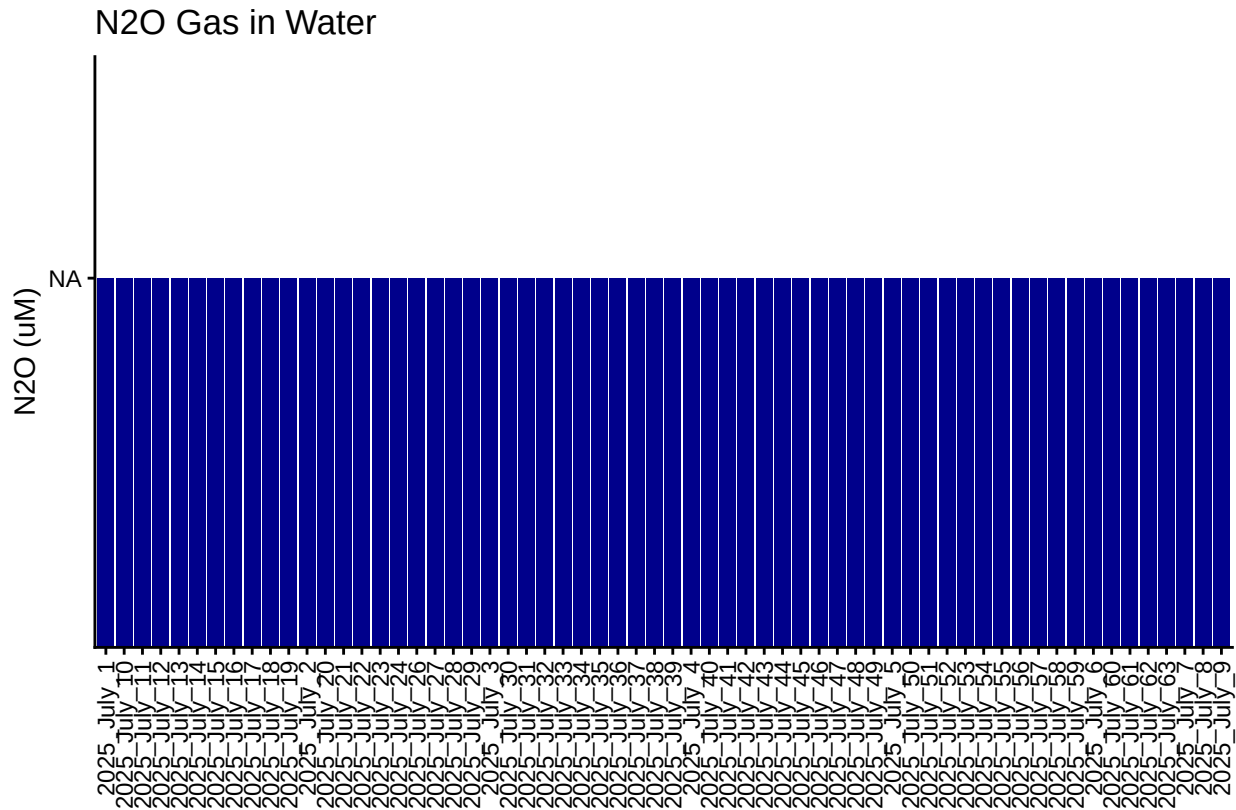
13 Calculate gas in water - CH₄



14 Calculate gas in water - CO2



15 Calculate gas in water - N2O



16 Merge samples with Metadata & check all samples are present

```
## Merge Samples with Metadata
```

```
## Some sample IDs are missing from data.
```

```
## [1] "2025_July_25"
```

16.1 Organize Data

```
##      Sample_ID  Type   Run_Date CH4_Area Conc_CH4      CH4_RTime_Flag CO2_Area
## 1 2025_July_1 Sample 2026-09-01    453      NA Correct - CH4 RTime    30406
## 2 2025_July_2 Sample 2026-09-01    331      NA Correct - CH4 RTime    40696
## 3 2025_July_3 Sample 2026-09-01    218      NA Correct - CH4 RTime    60972
## 4 2025_July_4 Sample 2026-09-01    590      NA Correct - CH4 RTime    21638
## 5 2025_July_5 Sample 2026-09-01    586      NA Correct - CH4 RTime    16359
## 6 2025_July_6 Sample 2026-09-01    438      NA Correct - CH4 RTime    51557
##      Conc_CO2      CO2_RTime_Flag N2O_Area Conc_N2O      N2O_RTime_Flag
## 1      NA Correct - CO2 RTime    551      NA Correct - N2O RTime
## 2      NA Correct - CO2 RTime    668      NA Correct - N2O RTime
```

```

## 3      NA Correct - CO2 RTime      604      NA Correct - N2O RTime
## 4      NA Correct - CO2 RTime      593      NA Correct - N2O RTime
## 5      NA Correct - CO2 RTime      657      NA Correct - N2O RTime
## 6      NA Correct - CO2 RTime      764      NA Correct - N2O RTime
##      FID_Peak_Flag ECD_Peak_Flag CH4_Conc_ppm      CH4_Flag CO2_Conc_ppm
## 1      Correct      Correct      1.1729763 Within Range      19524.55
## 2      Correct      Correct      0.7406847      BDL      26327.75
## 3      Correct      Correct      0.3402835      BDL      39733.17
## 4      Correct      Correct      1.6584185 Within Range      13727.61
## 5      Correct      Correct      1.6442450 Within Range      10237.41
## 6      Correct      Correct      1.1198257 Within Range      33508.47
##      CO2_Flag N2O_Conc_ppm N2O_Flag CH4_Conc_uM CO2_Conc_uM N2O_Conc_uM
## 1 Within Range      0.2053292      BDL      NA      NA      NA
## 2 Within Range      0.2812331      BDL      NA      NA      NA
## 3 Within Range      0.2397130      BDL      NA      NA      NA
## 4 Within Range      0.2325768      BDL      NA      NA      NA
## 5 Within Range      0.2740969      BDL      NA      NA      NA
## 6 Within Range      0.3435133      BDL      NA      NA      NA

##      Sample1 Sample_Year Sample_Month Sample_Day Sample_Time Time_Zone G_W
## 1      SGW      2025      July      17      1206      EDT      G
## 2      SGW      2025      July      17      1207      EDT      G
## 3      SGW      2025      July      17      1209      EDT      G
## 4      SGW      2025      July      17      1211      EDT      G
## 5      SGW      2025      July      17      1212      EDT      G
## 6      SGW      2025      July      17      1213      EDT      G
##      Sample_Plot Sample_Grid Sample_Depth Field_Notes Lab_Notes CH4_Conc_ppm
## 1      Control      C3      10      NA      NA      1.1729763
## 2      Control      C3      20      NA      NA      0.7406847
## 3      Control      C3      35      NA      NA      0.3402835
## 4      Control      C6      10      NA      NA      1.6442450
## 5      Control      C6      20      NA      NA      1.1198257
## 6      Control      C6      35      NA      NA      1.3005377
##      CH4_Conc_uM CH4_Conc_Flag CO2_Conc_ppm CO2_Conc_uM CO2_Conc_Flag N2O_Conc_ppm
## 1      NA      Within Range      19524.55      NA      Within Range      0.2053292
## 2      NA      BDL      26327.75      NA      Within Range      0.2812331
## 3      NA      BDL      39733.17      NA      Within Range      0.2397130
## 4      NA      Within Range      10237.41      NA      Within Range      0.2740969
## 5      NA      Within Range      33508.47      NA      Within Range      0.3435133
## 6      NA      Within Range      17619.78      NA      Within Range      0.2650143
##      N2O_Conc_uM N2O_Conc_Flag      QAQC_Flags Analysis_rundate Run_notes
## 1      NA      BDL N2O_R2 is Bad      2026-09-01
## 2      NA      BDL N2O_R2 is Bad      2026-09-01
## 3      NA      BDL N2O_R2 is Bad      2026-09-01
## 4      NA      BDL N2O_R2 is Bad      2026-09-01
## 5      NA      BDL N2O_R2 is Bad      2026-09-01
## 6      NA      BDL N2O_R2 is Bad      2026-09-01

```

16.2 Visualize Data CH4

```
## Visualize Data CH4
```

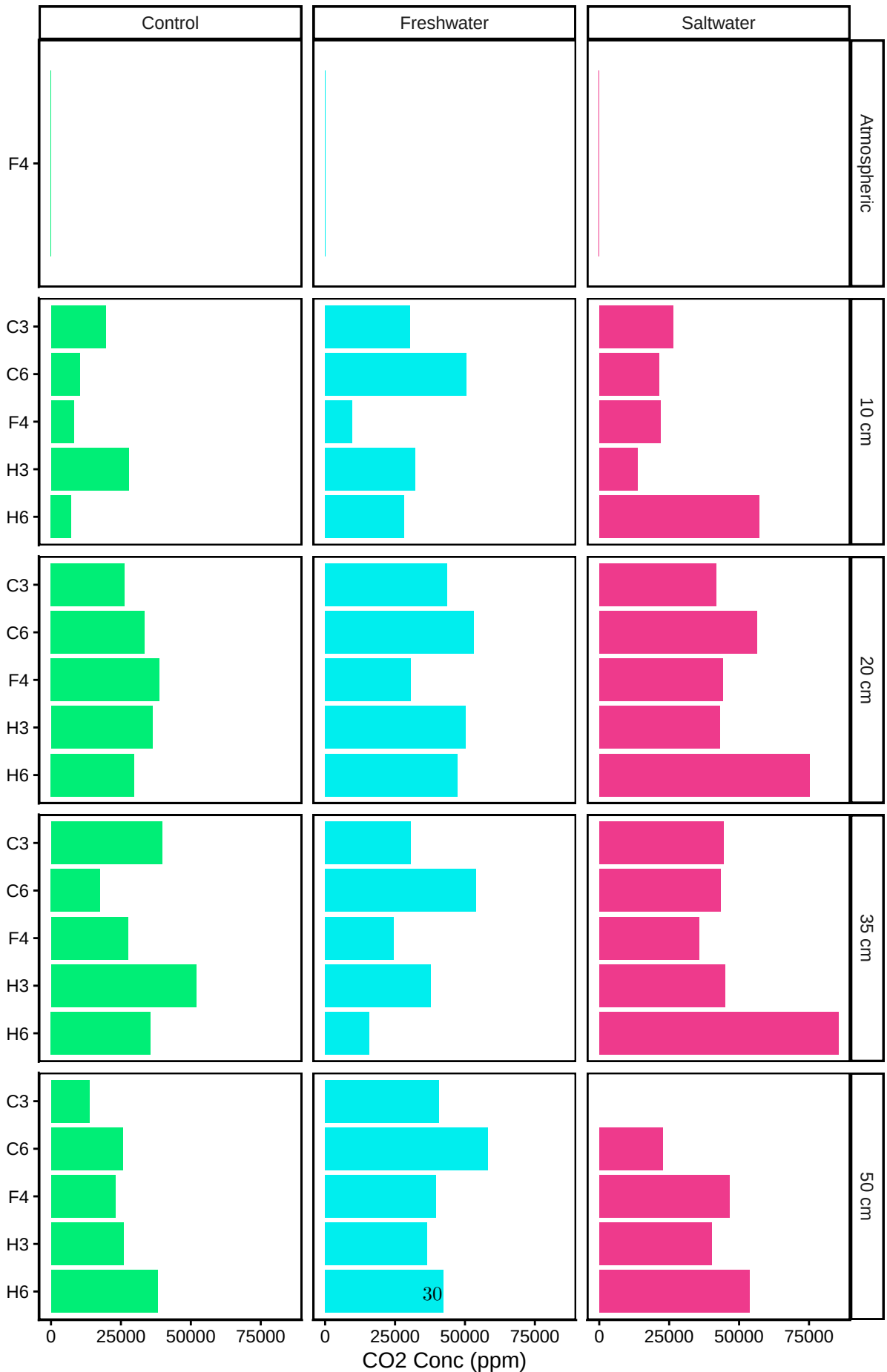
TEMPEST: Soil Gas Well CH4



16.3 Visualize Data CO2

```
## Visualize Data CO2
```

TEMPEST: Soil Gas Well CO2



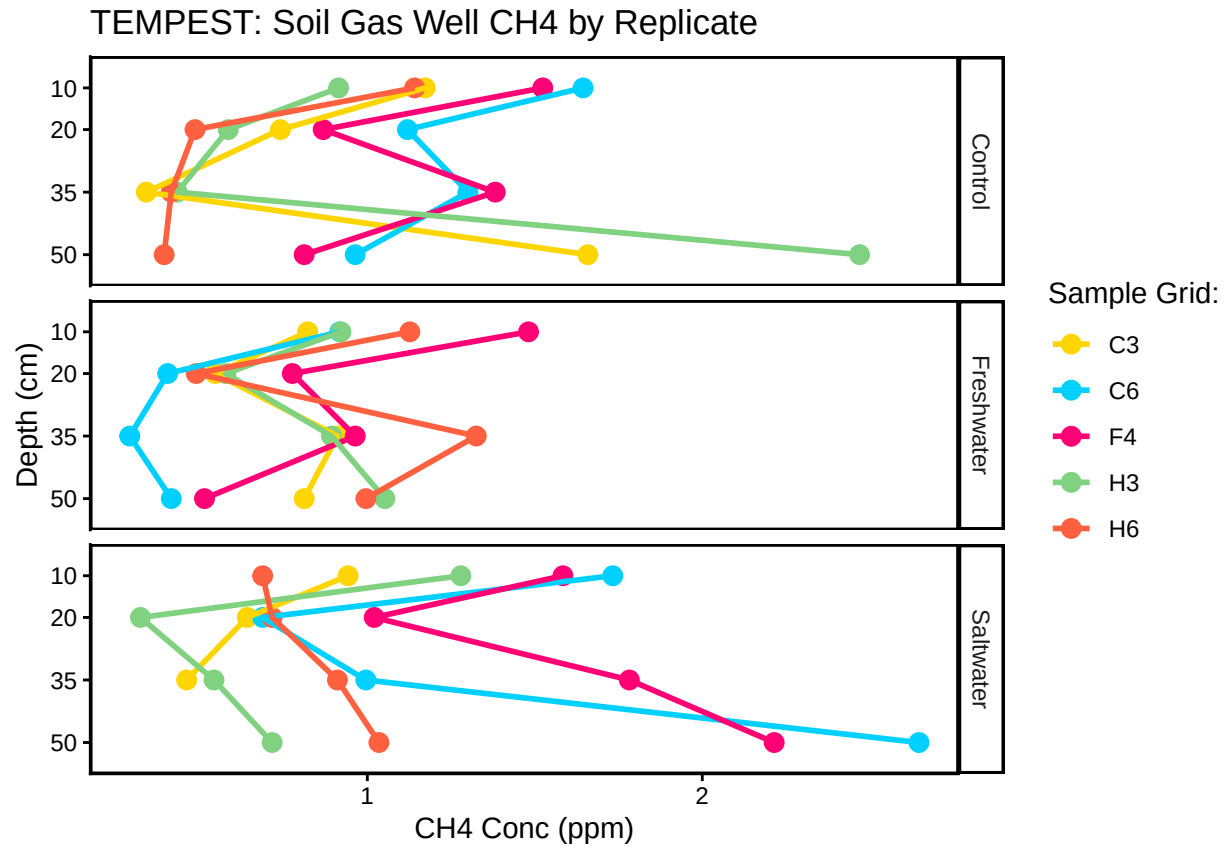
16.4 Visualize Data N2O

```
## Visualize Data N2O
```

TEMPEST: Soil Gas Well N2O

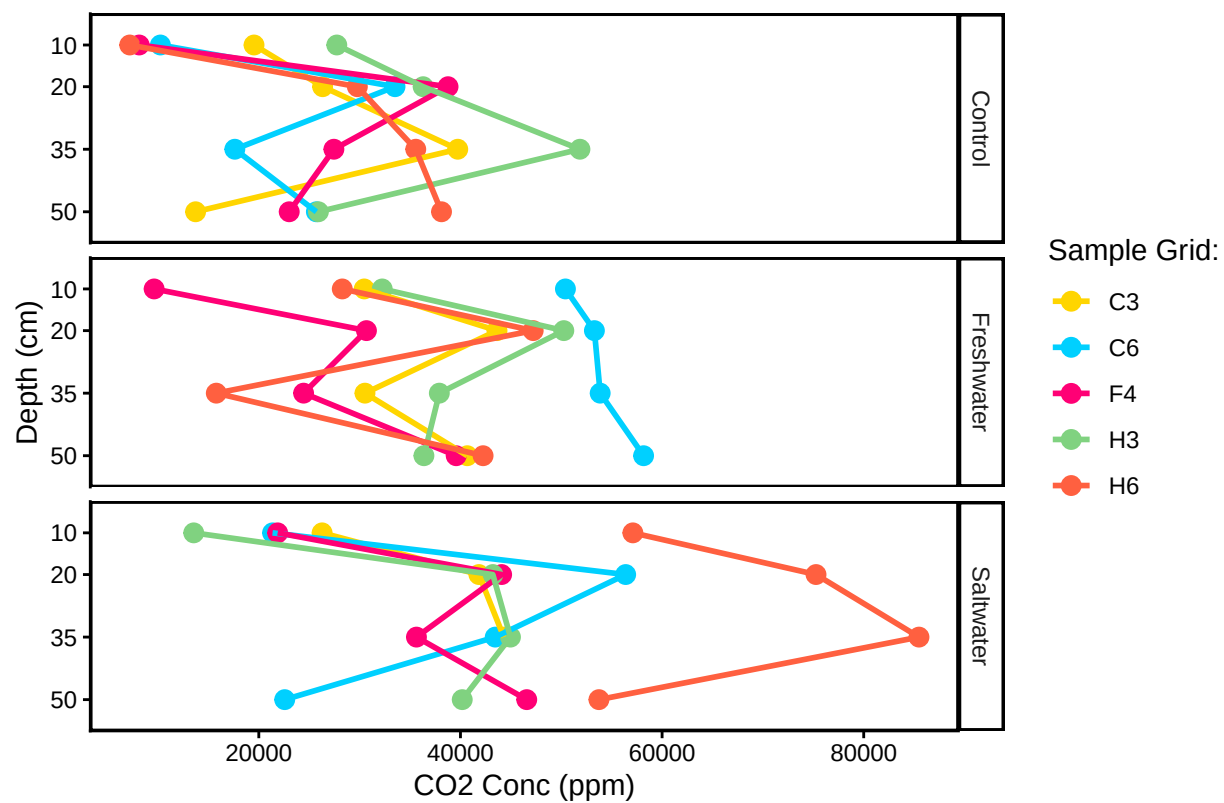


#Visualize Data CH4 by Replicate



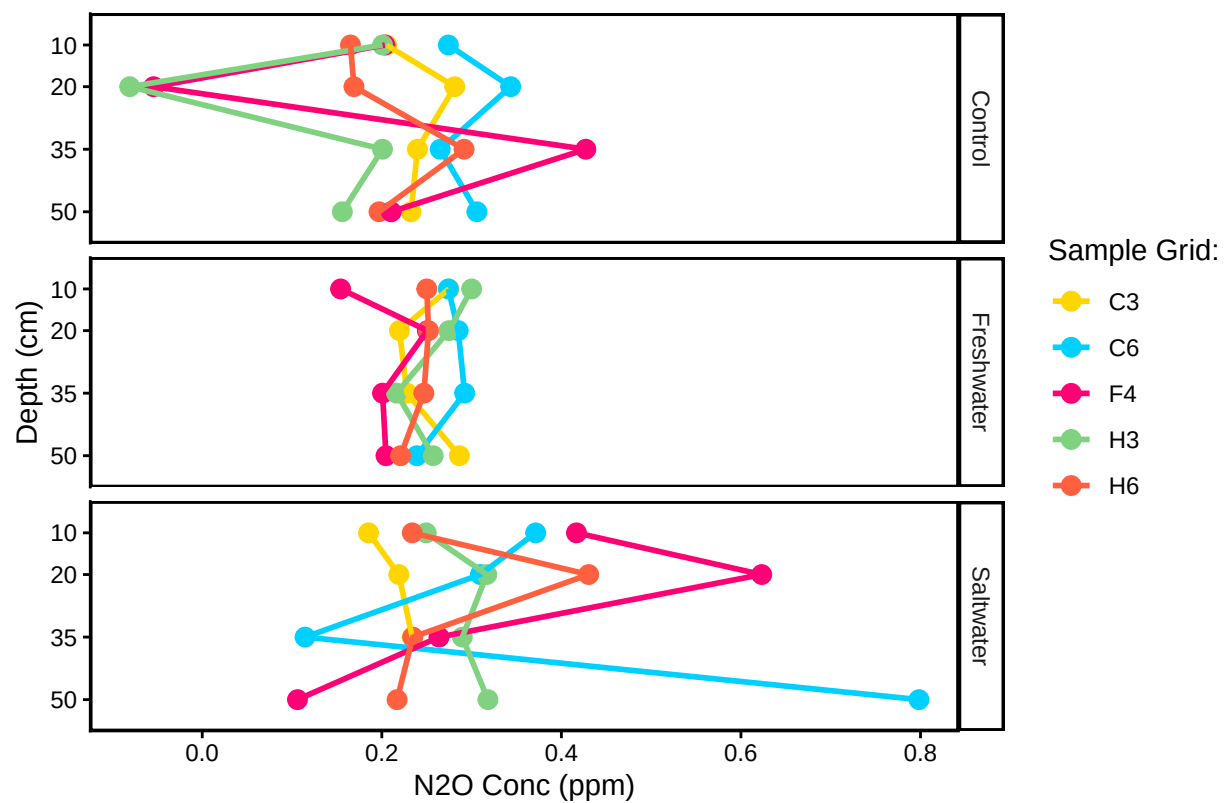
#Visualize Data CO2 by Replicate

TEMPEST: Soil Gas Well CO2 by Replicate



#Visualize Data N2O by Replicate

TEMPEST: Soil Gas Well N2O by Replicate



##Export Data

#end